

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could coordinated cortical activity support a unified experience and a minimal self? We develop a conditional field framework separating three requirements: coherence, compatibility and self-representation. Its physical hypothesis is that endogenous extracellular fields partly couple cortical processes, with changing extracellular geometry gating recovery after awakening. A noisy mean-field phase model predicts a square-root stationary onset and a von Mises concentration–coherence relation. Finite ramps delay that onset, and increased coupling, decreased diffusion and a matched common drive can give identical stationary phase distributions. A test therefore needs calibrated phase dynamics and measurements that distinguish the interaction mechanism. Beyond coherence, an explicit counterexample shows that phase agreement permits incompatible local states; a plasticity control preserves coherence while losing alignment with specified environmental structure. Compatible spatial measures glue uniquely, and a contracting self-prediction map has a unique fixed point, each relative to supplied local data and model choices. Lean 4 checks their composition under eight connecting hypotheses and supplied structures. We motivate the identification of the glued state with

conscious unity and of its accurate self-representation with a minimal self. The framework specifies what each requirement contributes and which additional measurements would make those identifications assessable.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Yet even successful coordination leaves a question: what makes the coordinated activity a description of one experienced situation, with a perspective on that situation?

We separate three requirements. *Coherence* concerns whether coupling sustains collective activity. *Compatibility* concerns whether local descriptions agree where their domains overlap, so they can belong to a single global description. *Self-representation* concerns whether that global state supports an accurate model of itself. These requirements organize the article. They are commitments of the construction developed here; they are not offered as requirements on every possible account of consciousness.

The distinction between timing and content is central. Imagine two local descriptions assigning different values to the same shared quantity. Making

19 their oscillations synchronous leaves that disagreement intact. The formal
20 construction exhibits exactly this separation: local spatial mass profiles at a
21 common phase can disagree on an overlap. Equal restrictions, rather than
22 equal phases, allow those profiles to glue into a unique global state. A self-
23 prediction map then asks a further question about that state: can a local
24 encoding reproduce the state of which it is a part?

25 Awakening gives the physical requirement a concrete setting. We hypoth-
26 esize that changing extracellular geometry slowly increases effective coupling
27 through a synchronization threshold. Under stationary branch tracking, a
28 noisy phase model fixes a square-root onset of coherence. The experiment
29 faces two distinct obstacles. Finite ramp speeds delay the response, so a
30 fitted onset exponent depends on the drive. Moreover, successive stationary
31 phase distributions can be identical under increasing coupling, decreasing
32 diffusion or a suitably matched common drive. Recovery must therefore be
33 measured in a way that identifies dynamics and distinguishes causes, rather
34 than inferred from the shape of a coherence trace alone.

35 The next requirement is representational. A plasticity control shows why
36 an adaptive coherent system need not acquire the environmental structure
37 it is scored against: a squared-drift objective rewards equalizing drift, which
38 in the chosen frequency-cluster model favors connections across the clusters
39 that define the template. This result motivates an explicit compatibility
40 condition and a measurement of that condition. It also locates the role
41 of thermodynamics: bounds on information and dissipation constrain an

42 implementation, while the content its variables represent requires additional
43 structure.

44 The final requirement connects unity to a minimal self. We propose that
45 a global state fixed by its own local encoding and readout is a formal cor-
46 relate of a state that is what it takes itself to be. The proposal earns its
47 interest by requiring the self-model to refer to the same state assembled from
48 local descriptions. Its psychological interpretation depends on identifying the
49 relevant descriptions, their domains and the encoding region. These choices
50 receive explicit attention in Section 5.

51 Lean 4 checks the mathematical composition underlying this account. Its
52 value is to keep different objects and claims distinct: a stationary density
53 and a trajectory, a dissipation bound and a learning rule, phase agreement
54 and agreement of local descriptions. The composition consumes eight named
55 connecting hypotheses and two structures with further physical content; a
56 witness assembled from distinct toy components satisfies them jointly. Sec-
57 tion 6 explains those obligations and the supporting thermodynamic results.
58 The empirical tests concern the physical coherence hypothesis; the interpre-
59 tation of a compatible, self-representing state as a conscious episode is argued
60 in Section 5.2. Figure 1 gives the structure of the argument.

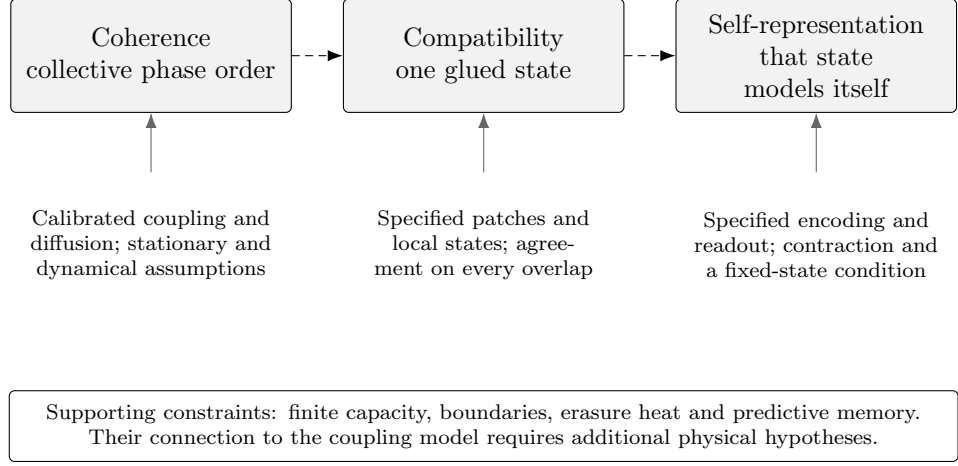


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by the formal composition.

2 Coherence: the coupling model

2.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are separate parameters.

68 For identical frequencies in their rotating frame, the stationary ansatz is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

69 where I_n are modified Bessel functions and ψ is the collective phase. Lean
70 verifies normalization, the circular order parameter and self-consistency for
71 this density. At $K \leq K_c = 2D$, the only nonnegative self-consistent order
72 parameter is zero. At $K > K_c$, exactly one positive solution coexists with
73 zero; it emerges continuously and increases strictly with K . Near threshold,

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

74 These results characterize the stationary branch (Fig. 2). Connecting it
75 to finite trajectories requires a dynamical mean-field limit and dynamical
76 selection, neither proved here. Propagation of chaos concerns the first of
77 these steps (Sznitman, 1991); exchange symmetry alone does not establish
78 independence, especially when realizations have an unpinned collective phase.

79 A spatial version replaces the finite sum by an integral against a kernel
80 $K(x, y)$. The substrate measure has total mass one, making the mean-field
81 strength a kernel average. Applying the scalar threshold to a heterogeneous
82 or signed kernel requires a justified reduction to that mean-field description.

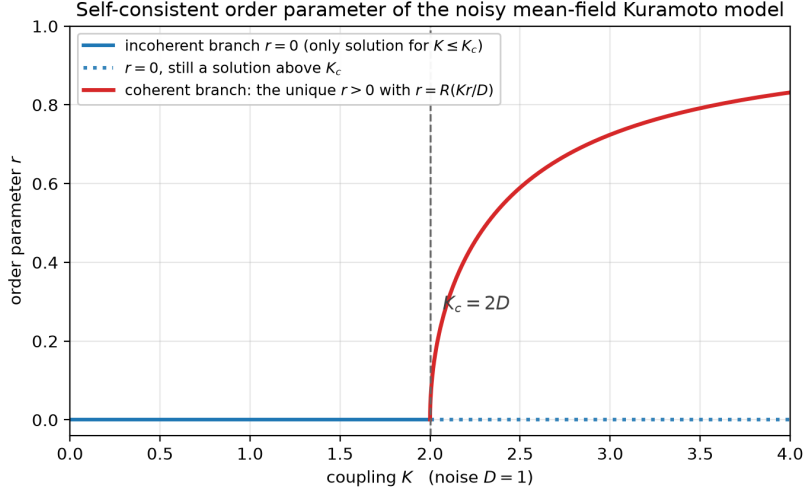


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

83 2.2 Physical identification and calibration

84 The cortical hypothesis identifies part of the effective coupling with endoge-
 85 nous extracellular electromagnetic interactions. Fields can modulate spike
 86 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field
 87 models have been proposed for coordinating distributed neural information
 88 (Pinotsis & Miller, 2022; McFadden, 2020). The quantitative question is
 89 whether that interaction supplies sufficient coupling relative to diffusion.
 90 Changes in phase organization under propofol (Bastos et al., 2021) moti-
 91 vate studying this regime, but do not measure the coupling itself.

92 The physical field predicate requires a jointly continuous kernel, a nor-

malized substrate, a mean-field strength equal to the spatial average, and no single site providing more than half that strength. Regular mesh refinement establishes convergence of a scalar coupling-energy quadrature; constructing the kernel and relating it to trajectories are separate modelling tasks. The formal witnesses supply toy instances of the predicate.

Calibration is essential because a spike-time shift is not a coupling rate. An estimate proportional to $N\Delta tf$, using population size, timing shift and oscillation frequency, is dimensionless. A conversion of the form

$$K_{\text{eff}} = \gamma N s E f \quad (4)$$

requires a rate γ , where s is the timing shift per unit field and E the field amplitude. This rate specifies how induced timing shifts persist and accumulate into phase drift. Supplemental Material explains why the other factors cannot determine it. The biological threshold test requires independent estimates of the interaction kernel, its aggregation, and phase diffusion in the same cortical regime.

3 Testing coherence through recovery after awakening

The proposed control variable is extracellular geometry. Sleep-associated changes in extracellular space and clearance (Xie et al., 2013) motivate a

111 coupling parameter that evolves more slowly than phase dynamics. The
 112 extracellular volume fraction is approximately $\alpha = 0.2$ at diffusion tortuosity
 113 $\lambda \approx 1.6$ (Syková & Nicholson, 2008); the cited sleep study reports a rise
 114 in α from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz cortical
 115 tissue conducts approximately resistively at 0.3–0.6 S m⁻¹ (Barbour, 2017).
 116 In an effective-medium description at fixed transmembrane current density,
 117 a larger extracellular volume fraction raises conductivity and reduces field
 118 amplitude.

119 This gives the coupling-gated hypothesis a directional prediction: contrac-
 120 tion of extracellular space on awakening raises E and, with the other factors
 121 in Eq. (4) held fixed, raises effective coupling. The proposed geometry–
 122 coupling relation and its role in human recovery both require measurement.
 123 The effective-medium argument supplies a sign under stated conditions; cal-
 124 ibration supplies the magnitude and the mapping to the scalar model.

125 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
 126 stationary branch, Eq. (3) gives

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

127 The crossing time satisfies $K(t_0) = K_c$. It is a prediction of latency only when
 128 coupling is measured independently of coherence. The subsequent shape
 129 and saturation depend on the later coupling trajectory. Critical slowing
 130 down restricts branch tracking near threshold, while finite populations have

131 fluctuating order below it. A measurable onset therefore needs an explicit
 132 criterion above that fluctuation floor.

133 **3.1 What a recovery distribution can identify**

134 Equation (2) gives a concentration–coherence relation with no fitted shape
 135 parameter. Testing it requires separate estimators: r from circular resultant
 136 length and a from density shape. Estimating a by inverting R from the
 137 measured r makes agreement automatic. Even a valid agreement tests a
 138 statistical description shared by the von Mises family; it does not determine
 139 the coupling medium.

140 There is a stronger limitation. Successive equilibrated phase distributions
 141 cannot, by themselves, distinguish three different recovery mechanisms. Let
 142 $q(t)$ be a prescribed ratio crossing 2, let $r(q)$ denote its stationary branch,
 143 and choose a fixed rate $D_0 > 0$. Increasing coupling with $K(t) = D_0 q(t)$
 144 and $D = D_0$ gives the same stationary density as fixed coupling K_0 with de-
 145 creasing diffusion $D(t) = K_0/q(t)$. Both have concentration $a = q(t)r(q(t))$.
 146 Uncoupled oscillators under a common prescribed drift $-h(t) \sin(\theta - \psi)$, with
 147 $h(t) = D_0 q(t)r(q(t))$ and $D = D_0$, have that same density. Their distribu-
 148 tions coincide window by window, even though their physical causes differ.
 149 Supplemental Material supplies the construction and its numerical feasibility
 150 check.

151 The missing observable is the physical clock. Resolved phase increments
 152 contain rate information: in the ideal continuous-path model, quadratic vari-

153 ation accumulates at rate $2D$, while conditional increment means constrain
 154 drift. This can separate coupling and diffusion under the specified observa-
 155 tion model. Distinguishing endogenous coupling from the matched common
 156 drive additionally requires a measured drive or a controlled perturbation, be-
 157 cause their mean-field drifts can coincide. Identifying the coupling as electro-
 158 magnetic then requires evidence specific to that interaction. This hierarchy
 159 determines the experiment: distribution, dynamics and physical mechanism
 160 are distinct targets.

161 **3.2 Finite ramps change the observable onset**

162 Finite simulations characterize how the stationary prediction can appear in
 163 a recovery measurement. Their numerical results are generated from saved
 164 summaries; the supplement gives parameters, estimators and further controls.
 165 The ramp study uses $N = 2000$, $D = 1$ and 32 replicas at four coupling
 166 speeds. Escape is the first of three consecutive saved samples at $r \geq 0.2$ after
 167 threshold.

168 Faster ramps delay escape (Fig. 3). Only 5/32 replicas escape within the
 169 fastest window. The three slower ramps have mean coupling delays 0.1947,
 170 0.0684 and 0.0253. Their fitted onset exponents run from 0.898 to 0.417,
 171 compared with 0.443 to 0.448 from the same estimator on the exact stationary
 172 branch at the same coupling grids. At the fastest speed the ensemble mean
 173 never reaches the declared fit window; the returned value 0.964 is a local
 174 log-slope of a delayed trajectory.

175 This is delayed passage through a noisy bifurcation (Berglund & Gentz,
 176 2002), with a direct consequence for the awakening test: the apparent expo-
 177 nent depends on the ramp and on the fitting window. A deviation from the
 178 square-root law constrains the adiabatic approximation only where branch
 179 tracking is independently justified. The sweep does not bracket the specified
 180 deviation-based critical ramp speed.

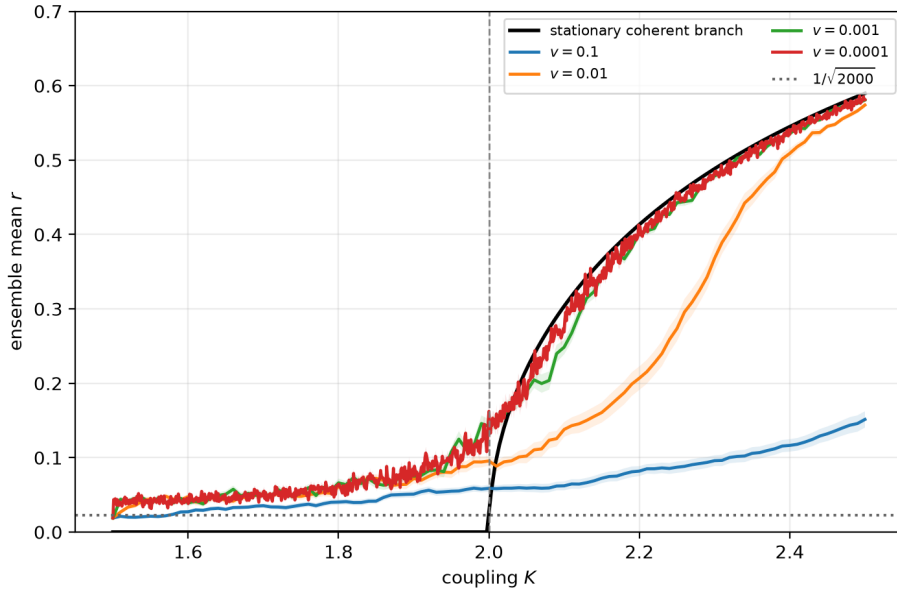


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

181 Additional controls test the approximations used by the physical model.
 182 Time-step refinement attributes most of the sampled discrepancy from the
 183 stationary order to integration bias. Population-size diagnostics show why

184 a shared random collective orientation can preserve unconditional pair cor-
 185 relation in the coherent regime. Spatial and signed-network controls exhibit
 186 coherence at specified total coupling, with the refined signed-network cross-
 187 ing bracket $K_{\text{eff}}/D \in [1.958, 2.009]$ still containing 2. These controls delimit
 188 finite regimes and numerical resolution; the supplement reports their full
 189 evidence. Calibrating the cortical coupling remains the experimental task.

190 **3.3 What the estimators produce on EEG**

191 Public propofol-sedation EEG (Bajwa et al., 2025) provides an exploratory
 192 exercise of the concentration and coherence estimators. The implemented
 193 analysis pools channels over 100-ms windows, estimating a spatiotemporal
 194 phase distribution. Bipolar and common-reference montages produce differ-
 195 ent residuals. The bipolar results are compatible with the Bessel relation,
 196 but the concentration range is too narrow to distinguish it from its linear
 197 approximation at the observed scatter.

198 An informative recording needs a broader concentration range, estima-
 199 tor calibration that accounts for dependence, and an observation model for
 200 source mixing, referencing and temporal pooling. The sleep-recovery hypoth-
 201 esis remains untested by these sedation recordings.

202 **3.4 A protocol for the awakening experiment**

203 The analysis defines six requirements for the physical test.

- 204 **1. Coupling and diffusion in the same units.** Estimate K and D
205 from resolved phase dynamics in the same cortical regime, separating
206 diffusion, frequency heterogeneity and observation noise. A measured
207 drive or controlled perturbation is additionally needed to distinguish
208 endogenous coupling from the matched common-drive alternative.

- 209 **2. Geometry calibrated to coupling.** Establish the relation between
210 an extracellular-geometry observable and effective coupling, with un-
211 certainty. A diffusion-MRI or astrocytic proxy for volume fraction re-
212 quires its own validation; the conductivity argument supplies only the
213 expected sign.

- 214 **3. Adequate phase measurements.** For the proposed spatial estima-
215 tor, use at least 100 sites per estimate, r from resultant length, and
216 a from unweighted regression of log density on the cosine of centered
217 phase. This numerical minimum needs null calibration at the actual
218 sample count, concentration range and dependence. Temporal pooling
219 changes the estimand.

- 220 **4. An independent recovery endpoint.** Measure behavioral recovery
221 separately from the phase trace.

- 222 **5. A predeclared onset criterion.** Fix a coherence threshold above the
223 finite-size floor and account for finite ramp speed before interpreting
224 an onset exponent.

225 **6. Matched model comparison.** Compare coupling-gated recovery
 226 with diffusion change, common drive and alternative slow-recovery
 227 models under the same observation and filtering assumptions. Station-
 228 ary distributions alone cannot resolve the equivalence in Section 3.1.

229 Failure of the concentration–coherence relation over an informative range
 230 would constrain the phase-density model. Failure of the calibrated geometry–
 231 coupling association would constrain the proposed physical gate. A com-
 232 peting dynamical explanation that better predicts independently measured
 233 recovery would count against its role in awakening. These tests address the
 234 first requirement of the framework. Even a favorable outcome leaves the
 235 question of what the coordinated activity represents.

236 **4 Compatibility: from coordinated activity** 237 **to a global description**

238 **4.1 Why common timing is insufficient**

239 Consider overlapping patches U and V , each carrying a local description.
 240 Compatibility requires that restricting either description to $U \cap V$ gives the
 241 same object. Agreement concerns the shared domain; descriptions on distinct
 242 parts can remain different. For the finite spatial measures used here, equal
 243 overlap restrictions give a unique measure on the covered domain. This is
 244 the gluing property formalized by a sheaf.

245 The formal counterexample makes the additional requirement visible. Ev-
246 ery patch has the same oscillator phase, yet two local mass profiles assign
247 different densities to a shared site. They cannot be restrictions of a single
248 global measure. Phase order therefore cannot establish compatibility. Con-
249 tinuity of a physical field does not repair disagreement in the descriptions
250 associated with it.

251 For a theory of experience, the attraction of gluing is that local contents
252 contribute to one description without each region having to carry that entire
253 description. Its mathematical realization here is deliberately specific: the
254 objects are spatial measures. A decoder assigning probability laws to per-
255 ceptual contents needs a further extension model, because matching overlap
256 marginals need not determine a unique global joint distribution (Abramsky &
257 Brandenburger, 2011). The passage from spatial profiles to content-bearing
258 neural descriptions is consequently a substantive modelling commitment.

259 **4.2 Coherent adaptation can lose environmental align-** 260 **ment**

261 An adaptive coupling rule might appear to supply that missing representa-
262 tional structure. The squared-drift control shows why the objective mat-
263 ters. Symmetric coupling conserves mean drift: pairwise sine terms cancel
264 when summed over nodes. Minimizing the sum of squared drifts at that
265 fixed mean therefore rewards equalizing drift across nodes. In the chosen

frequency-cluster model, this favors coupling between unlike-frequency clusters, whereas the diagnostic template rewards coupling within like-frequency clusters. The objective and the scored structure pull in different directions.

For fixed phases, the continuum objective is

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

The formalization constructs the bounded drift operator A and its adjoint, obtains $\nabla\sigma = (2/D)A^*(AK + \omega)$, and proves descent for the specified gradient flow. This is a squared-drift functional; identifying thermodynamic entropy production would require probability-current information.

The finite control co-evolves noisy phases and symmetric, nonnegative couplings at fixed total resource. Across three seeds it compares gradient updates, random updates scaled to each arm’s own gradient norm, randomly relabelled gradient updates, and a frozen kernel. The frequency-cluster template is a diagnostic covariance of shared latent signals, not a training input. Figure 4 shows the decisive dissociation: second-half order remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135. The relabelled arm reaches overlapping deformation and coherence ranges but a ratio of 0.7290–1.2935, above the gradient arm’s range. Supplementary controls compare partitions blind to frequency and distinguish alignment from growth of the kernel norm.

The objective reduction is transient. Immediately after updates, the

286 gradient arm removes a fraction 0.6773–0.6914 of the headroom above the
 287 conserved-drive floor. Averaging over complete update intervals removes that
 288 apparent sustained gain. A prespecified grid of 6 rate–cadence combinations
 289 contains no case meeting the declared 5% sustained-reduction criterion while
 290 retaining coherence and losing alignment. The evidence therefore concerns
 291 coherent adaptation away from a specified structure, not sustained minimiza-
 292 tion of a thermodynamic cost.

293 This control is not a test of overlap compatibility itself: kernel alignment
 294 and agreement of local descriptions are different observables. Its role is to
 295 block a proposed shortcut to compatible representations through coherent,
 296 objective-descending dynamics. The structure learned by slow variables de-
 297 pends on their objective and how environmental information enters it. A
 298 thermodynamically specified learning model can include data in that objec-
 299 tive (Corberi et al., 2026); the dissipation bound alone does not supply such
 300 a mechanism.

301 **4.3 What must be specified and measured**

302 The framework supplies a cover of local states as an argument, with overlap
 303 compatibility among its fields. Its proposed physical bridge, E78, connects
 304 supercritical coherence to relaxation toward the equilibrium of that cover.
 305 For deterministic positive coupling, convergence to a potential minimum is
 306 proved under an explicit initial basin condition. Connecting that relaxation
 307 to the noisy stationary order parameter requires E78. Compatibility itself

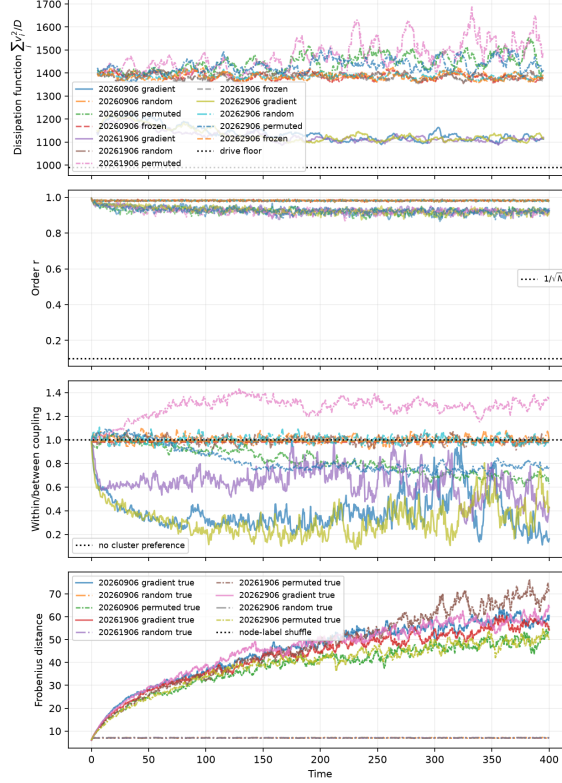


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

308 is supplied by the cover, so the construction does not explain how initially
309 inconsistent representations become consistent.

310 A neural test must first identify the domains and objects whose agreement
311 is at issue. Two decoders should estimate a shared content on overlapping
312 domains, with restriction maps specified before inspecting their agreement.
313 Their discrepancy needs a null under decoding noise, and the procedure must
314 distinguish deliberately constructed compatible and incompatible examples.
315 A common prior can make separate decoders agree artificially; region-specific
316 bias can make them disagree artificially. The supplement develops these re-
317 quirements. An estimator built from phase agreement would miss the prop-
318 erty the counterexample separates.

319 Choosing a cover is also part of choosing a candidate subject. Trivial
320 descriptions can agree while carrying little content, and a suitably chosen
321 partition can conceal disagreement. The scientifically useful choice must
322 therefore be constrained independently by what local activity encodes and
323 which shared quantities can be compared. The uniqueness theorem then
324 answers a precise question about that choice: whether the supplied local
325 descriptions determine one global state. It leaves the empirical selection of
326 domains and descriptions to the model.

327 5 Self-representation: a state that models it- 328 self

329 5.1 The reflexive construction

330 Compatibility yields a global state. Reflexivity adds a relation between that
331 state and an encoding within it. Equip global sections with a complete
332 metric. Under the additional restriction-resonance condition, the encoding
333 of a section s is its restriction to a region U , and the map takes the form

$$F(s) = \text{readout}(s|_U). \quad (7)$$

334 The same region and readout are used across the model's state space. If F
335 contracts and fixes the section s_* supplied by the cover, Banach's theorem
336 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
337 plies the contraction law and $F(s_*) = s_*$ for the boundary's encoding-readout
338 composite. Identifying that encoding with restriction is a separate resonance
339 condition; E89 does not supply it. The supercritical order parameter pro-
340 vides an inequality used by the specified contraction bound; constructing the
341 encoding and readout remains a separate obligation.

342 The point of requiring this particular fixed point is that the self-model
343 refers to the state formed from the local descriptions. A fixed point of an
344 unrelated map would leave that relationship unexplained. The equality ex-
345 presses accuracy at the level of the modelled state: the encoding reads back

346 the state that contains it. The term *predictive* names this reconstruction
347 map; identifying its operation with a temporal neural prediction requires a
348 dynamical interpretation.

349 **5.2 Why connect these properties to unity and a min-** 350 **imal self?**

351 We propose two identifications. The section glued from compatible local
352 states is the formal correlate of the unity of a conscious episode. Its ac-
353 curate self-representation is the formal correlate of a minimal self. These
354 are philosophical commitments, motivated by the relations the construction
355 expresses.

356 The unity proposal starts from a distinction between having several con-
357 tents and having several experiences. A globally coherent description can
358 contain many different local contents because compatibility only constrains
359 their shared domains. Gluing makes their contribution to one state explicit.
360 It also demands care about what is encoded. An episode can include uncer-
361 tainty or conflicting judgments. The relevant question is whether its local
362 descriptions can consistently represent that uncertainty or conflict, rather
363 than whether every represented proposition is true or every judgment agrees.
364 Applying the proposal requires a content model that preserves this distinc-
365 tion.

366 The self proposal addresses a different feature: the global state has a

367 representation of its own condition within the system that realizes it. A
368 region encoding the surrounding world alone would not meet this require-
369 ment. Restriction followed by readout supplies a precise version of internal
370 self-reference, and the fixed-state condition expresses accuracy. In this lim-
371 ited sense the state is what it takes itself to be. Nothing in that description
372 requires a verbal report, an autobiographical narrative or explicit reflection.

373 Accuracy is a demanding idealization. The construction concerns the
374 variables represented by s_* , not an exhaustive internal copy of every mi-
375 croscopic physical fact. A cortical realization would have to specify those
376 variables and justify the fidelity needed for the identification. Approximate
377 self-models, errors and changing states may require a weaker condition than
378 an exact fixed point. Such a condition is a research question rather than a
379 consequence of the theorem.

380 The relation between the two clauses is itself a commitment. Gluing alone
381 does not require self-representation. The full identification proposes that
382 unified conscious episodes satisfy both clauses, with the self-model about
383 the very state whose unity gluing expresses. The necessity of reflexivity is
384 a philosophical hypothesis, open to dissociation. Unified experience with-
385 out the proposed self-encoding would count against that requirement, while
386 leaving the compatibility proposal assessable on its own.

387 5.3 Which cover and which self-model?

388 Both uniqueness results are conditional on their objects. Gluing gives one
389 global extension of specified compatible local data; Banach gives one fixed
390 point of a specified contracting map. These solve ambiguity within a chosen
391 construction. They do not choose between different covers, different encod-
392 ings or different candidate subjects. Integrated information theory makes
393 exclusion an explicit part of its account (Tononi et al., 2016); the present
394 uniqueness results do not settle that selection problem.

395 The relevant cover and map must be tied to neural organization by evi-
396 dence independent of the desired conclusion. Local descriptions should pre-
397 dict the contents attributed to their domains; their restrictions should pre-
398 serve meaning on shared domains; and an encoding region should carry in-
399 formation about the proposed global state that its readout can recover across
400 an independently specified range of states. These are criteria for a biological
401 model, beyond existence of the mathematical objects.

402 This matters because contraction and a fixed point are easy to obtain in an
403 uninformative construction. A constant readout returning one predetermined
404 state would satisfy those two conditions while explaining little about self-
405 representation. Local encoding and the restriction-resonance condition make
406 the proposed relation explicit, but its psychological significance still depends
407 on a demonstrated relation to the state. An intervention on an encoding
408 should have consequences predicted by that relation. The theory needs this
409 empirical identification as much as it needs calibrated coupling.

410 5.4 Dissociations that would bear on the identification

411 A test of the full commitment must measure its clauses separately: effective
412 coupling relative to diffusion, overlap compatibility, and the accuracy of the
413 proposed self-encoding, alongside an independent endpoint for experience.
414 A case meeting all three physical and representational conditions with evi-
415 dence against a unified conscious episode would challenge their sufficiency.
416 Unified experience with demonstrably incompatible local descriptions would
417 challenge the unity identification. Unified experience with compatibility but
418 without the specified self-encoding would challenge the reflexive requirement.

419 Anesthetic emergence, dreaming and dissociative states are candidate set-
420 tings because the framework asks whether coordination, unified content and
421 self-representation can separate. Interpreting such evidence requires a de-
422 clared relation between the experience endpoint and report; absence of report
423 alone cannot define absence of an episode. The compatibility observable in
424 the supplement is a specification, and the corresponding neural self-encoding
425 has yet to be identified. These are the main obstacles to testing the inter-
426 pretive commitment.

427 The proposal concerns which state is conscious. It supplies no explanation
428 of why satisfying these relations should be accompanied by experience. Nor
429 do its gluing and fixed-point conditions select an electromagnetic substrate:
430 the cortical field commitment resides in the physical identification and cali-
431 bration. Their multiple realizability leaves the status of other biological and
432 artificial implementations open.

433 6 Thermodynamic constraints and the con- 434 ditional composition

435 The three requirements can be stated beginning with the phase model. Fi-
436 nite capacity, symmetry breaking and erasure are not needed to derive its
437 stationary threshold. Their role in the broader framework is to constrain a
438 proposed physical implementation: what it can retain, where its degrees of
439 freedom are bounded, and what its updates dissipate. The formal composi-
440 tion asks whether such an implementation can be connected consistently to
441 the coherent, compatible and reflexive state just described. It is one proposed
442 architecture, not a necessary route from thermodynamics to consciousness.

443 6.1 Capacity and predictive memory constrain an im- 444 plementation

445 For a finite state space X , Shannon entropy satisfies $H(p) \leq \log |X|$, with
446 equality only for the uniform law. A stream with more possible histories than
447 $|X|$ cannot be encoded injectively in that register. Separately, a continuous
448 field on a connected substrate reaching distinct components of a disconnected
449 vacuum manifold must leave that manifold somewhere (Kibble, 1976). The
450 finite register and the field boundary are independent starting objects. Iden-
451 tifying them as parts of one physical system is an assumption.

452 A non-surjective update on a finite register is non-injective. Under the

453 specified system–bath model, erasure carries a Landauer heat cost (Landauer,
 454 1961); connecting logical many-to-one processing to physical heat requires
 455 implementation assumptions (Norton, 2005; Shenker, 2000). For a state X_t
 456 driven by an environmental signal S_t , the predictive dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (8)$$

457 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, the bracket is
 458 nonnegative (Still et al., 2012). The bound limits nonpredictive memory at
 459 a given dissipation budget. Deriving it from the register’s Landauer bound
 460 requires the nonpredictive information to be bounded by the entropy erased,
 461 with the same thermal scale and heat budget.

462 This constrains memory without specifying its content or learning rule.
 463 Zero retained information can satisfy the bound, so reducing its right-hand
 464 side does not require better representation. A bound $D_{KL} \leq \Delta t \sigma$ forces
 465 convergence of D_{KL} to zero only when its right-hand side tends to zero.
 466 Counting capacity, erased entropy and predictive mutual information have
 467 different roles (Rosas et al., 2026). The plasticity control makes the conse-
 468 quence concrete: an objective must be related to the structure a system is
 469 expected to learn.

470 6.2 What the composition establishes

471 Table 1 lists the eight implications accepted by the theorem `chain`. Its node
472 predicates concern finite capacity, a symmetry-breaking boundary, erasure
473 heat, a predictive dissipation bound, a continuum coupling limit, a phys-
474 ical field realization, coherent order, a glued section and a reflexive fixed
475 point. The ordering specifies the theorem’s arguments. In particular, the
476 implication from finite capacity to a boundary is supplied independently;
477 the ordering does not turn it into a deduction.

478 Two structures also enter the theorem: a thermodynamic cover, including
479 overlap compatibility, and a reflexive boundary specifying the local encoding
480 and readout. Given these structures and the connecting hypotheses, `chain`
481 concludes `UnifiedSelf`: the particular section glued on that cover is the
482 unique fixed point of self-prediction. The theorem does not require the sep-
483 arate restriction-resonance predicate; that condition is needed to interpret
484 its encoding as the restriction in Eq. (7). The conclusion binds the end-
485 point to the supplied objects; its interpretation as a conscious state is the
486 commitment of Section 5.2.

487 The formal development includes a joint witness assembled from a double
488 well, a register, a uniform-grid mesh and a three-site cortex. It establishes
489 that the hypotheses can hold together, while the components remain distinct
490 toy systems. Counterexamples such as phase agreement without compatible
491 profiles expose the work individual assumptions do. The development de-
492 clares no additional Lean axioms beyond the foundational axioms used by

Connection	Status	Required assumption
E12	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity and the boundary are independent roots.
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34	Bridge assumption	Erase supplies a predictive structure with the same thermal scale and heat budget, and nonzero wasted memory.
E45	Modelling assumption	The predictive dissipation bound places coupling energies in a convergent coarse-graining regime.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence drives a supplied cover to its relaxed equilibrium at coupling at least the named mean-field value.
E89	Modelling assumption	A contracting map mediated by local self-encoding fixes the section glued on that same cover.

Table 1: Connecting hypotheses of the formal composition. The cover’s compatibility is supplied as a structure field. Identifying the boundary’s encoding with restriction requires a separate resonance condition. The mathematical objects and node-level results are mapped in Supplemental Material, Table S1.

493 its libraries; physical content is carried by hypotheses and model fields. This
494 makes the conditional construction checkable without treating a biological
495 premise as a proved fact.

496 **7 Discussion**

497 This construction separates three proposed requirements for conscious unity:
498 collective timing, compatible local descriptions and an accurate representa-
499 tion of the resulting global state. Each answers a different question. Coupling
500 can sustain coherence; gluing gives one state from specified compatible de-
501 scriptions; and reflexivity relates that state to an encoding within itself.

502 The physical proposal is a geometry-mediated increase in effective cou-
503 pling during awakening. Its most useful experimental consequence is a hi-
504 erarchy of measurements. Concentration and coherence test a phase-density
505 description. Resolved increments add rate information. A measured drive or
506 selective perturbation is needed to distinguish interaction mechanisms. Fi-
507 nite ramps determine how the transition appears, so a fitted onset exponent
508 cannot substitute for those measurements. Calibration and identifiability
509 govern what a recovery experiment can conclude.

510 The representational proposal begins where a synchronization account
511 runs out of explanatory content. Equal phases permit disagreement on over-
512 laps, and the specified plasticity rule can sustain coherence while moving
513 away from environmental alignment. The framework therefore assigns com-

514 patibility its own mathematical condition and measurement task. A biologi-
515 cal account must explain which local descriptions carry relevant content and
516 how their agreement is maintained. Thermodynamic bounds constrain this
517 account without supplying the missing content model.

518 The reflexive proposal adds an internal representation of the same global
519 state. Its appeal is the explicit relationship between unity and self-reference;
520 its cost is the need to identify a cover and encoding independently, justify
521 the accuracy idealization and test whether unified experience requires that
522 relation. These are substantive commitments left open by uniqueness relative
523 to fixed mathematical objects.

524 Compared with integrated information, global-workspace and predictive-
525 processing accounts (Tononi et al., 2016; Baars, 1997; Dehaene et al., 2014;
526 Friston, 2010), the framework places a candidate physical interaction beside
527 a compatibility condition and a local self-encoding. Its contribution is a con-
528 ditional connection between them, with concrete examples of where simpler
529 inferences fail. The next decisive evidence would couple a calibrated recov-
530 ery experiment to independently validated measures of content compatibility
531 and self-representation. Together those measurements would ask whether the
532 cortical state that becomes coordinated is also the state that fits together and
533 represents itself.

534 **Data and code availability**

535 The Lean development, simulation scripts, tests and saved numerical sum-
536 maries are maintained in the manuscript source repository. Publication sim-
537 ulation values are regenerated from saved summaries without rerunning pro-
538 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
539 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
540 in Supplemental Material. The source repository contains the reproducible
541 analysis and does not redistribute the raw recordings.

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545 **Use of AI tools**

546 AI assistance to this work was substantial and is not separable by task.
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549 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
550 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
551 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
552 they contributed to the Lean development, the simulation and analysis code,

553 and the drafting and revision of both documents. Because the tools were
554 used in combination and in alternation, no section, proof, figure or result is
555 attributable to any one of them, and none is claimed to be. The author is
556 responsible for all content, including everything a tool produced. Machine
557 checking establishes the stated Lean results under their hypotheses; it does
558 not validate biological interpretations or replace scientific review of generated
559 code and prose.

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